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11 March 1981
SUPERSEDING
MIL-V-17462A(SHIPS)
11 April 1961
(See 6.7)

MILITARY SPECIFICATION

VALVE, SAFETY: BOILER (SHIPBOARD USE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 1-1/2-inch and 2-1/2-inch boiler safety valves for steam drum and superheater service at working pressures of 600 and 1500 pounds per square inch (lb/in^2).

1.2 Classification. Safety valves shall be of the following types, classes, and sizes, as specified (see 6.2.1):

Type I - Flanged, direct spring loaded, pop type.
Type II - Flanged, steam drum pilot actuator.
Type III - Flanged, superheater outlet pilot actuated.
Class 1 - 600 lb/in^2 service.
Class 2 - 1500 lb/in^2 service.
Size 1 - 1-1/2-inch.
Size 2 - 2-1/2-inch.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.
MIL-R-196 - Repair Parts, Accessories, and Kits, Mechanical; Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4820

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MILITARY (Continued)

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-S-1222 - Studs, Bolts, Hex Cap Screws, and Nuts.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.

DRAWING

NAVAL SEA SYSTEMS COMMAND

NAVSHIPS 5000-S5103-841622 - Valves, Boiler Safety, 300, 600, and 1500 Psi Nominal Pressure, Dimensional Standard for.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 182 - Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High Temperature Service.
A 216 - Carbon-Steel Castings Suitable for Fusion Welding for High-Temperature Service.
A 217 - Martensitic Stainless Steel and Alloy Steel Castings for Pressure-Containing Parts Suitable for High Temperature Service.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

B16.5 - Steel Pipe Flanges, Flanged Valves, and Fittings.

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(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. Valves furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at time set for opening of bids (see 4.3 and 6.3).

3.2 Material. Materials used in the construction of safety valves shall be of the quality best suited for the purpose intended. Material of bodies, bonnets, and bolt-studs or studs and nuts for inlet flanges and bonnets shall be as specified herein. The suitability of the material of all other parts shall be demonstrated by satisfactory performance of the valves when tested as specified herein and in service.

3.2.1 Bodies. Bodies of all safety valves shall be fabricated from the applicable grade or composition of material as follows:

(a) For maximum steam drum pressure of 750 lb/in^2 (class 1), type I or type II, material shall be of cast steel conforming to ASTM A 216.
(b) For maximum steam drum pressure of 1600 lb/in^2 (class 2), or maximum steam temperature of 875°F, material shall be of carbon molybdenum steel conforming to grade F1 of ASTM A 182 or grade WC1 of ASTM A 217; or of chromium molybdenum steel conforming to grades WC6 or WC9 of ASTM A 217 or grades F11 or F22 of ASTM A 182.
(c) For superheated steam temperatures between 875°F and 975°F (at any pressure), material shall be of chromium molybdenum steel (weldable grade) conforming to grades WC6 or WC9 of ASTM A 217 or to grades F11 or F22 of ASTM A 182, or better.
(d) For superheated steam temperatures exceeding 975°F, material shall be of chromium molybdenum alloy steel or corrosion-resisting steel of a composition satisfactory to the command or agency concerned.

3.2.2 Bonnets. Bonnet material shall be in accordance with the following:

(a) Steam drum valve bonnets shall be of the same material as specified in 3.2.1.
(b) Materials for superheater outlet spring loaded valve bonnets shall have a low mean coefficient of thermal expansion (less than 7.95 x 10^{-6} in/in/°F). Where required, use of cooling spools or other means to reduce thermal effects may be used.

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3.2.3 Bolts, studs, and nuts. Bolts, studs, and nuts shall conform to MIL-S-1222. In general, the following shall apply:

(a) Up to 650°F, use material grade 5 for bolting.
(b) Up to 850°F, use material grade B7 for bolting.
(c) Up to 1000°F, use material grade B16 for bolting and 2H for nuts.

3.2.4 Nozzles, discs, and adjusting rings. Nozzles, discs, and adjusting rings shall be made from corrosion-resisting materials suitable for the service intended.

3.2.5 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 General construction and design.

3.3.1 Safety valves shall be constructed and shall operate so that shocks detrimental to the valves or to the boiler are minimized when lifting or closing.

3.3.1.1 Valves shall withstand maximum expected loading in any direction which shall include the load which may be applied by the escape piping (i.e., expansion and reactive force of the steam with the valve blowing) while remaining within valve pop, repeatability, performance, or leakage limits.

3.3.1.2 A temporary metal tag shall be affixed to the bolt holes of each valve exhaust flange, engraved with the following statement: "Minimize valve exhaust flange loading by adjusting exhaust piping and expansion joints for no-load condition when the boiler is hot and boiler expansion has stabilized."

3.3.2 The blowdown of all safety valves shall be adjustable over a range of from 2-1/2 to 6 percent of the design popping pressure. Clear and complete adjustment instructions shall be included in the manufacturer's maintenance manual.

3.3.3 The design of the bonnet, blowdown adjusting device, and valve disc shall be such as to minimize back pressure in the valve body acting upon the valve disc.

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3.3.3.1 It shall be possible to remove or install the lifting gear bracket assembly without disturbing the tightness of the bonnet. To accomplish this, the bonnet studs which receive the lifting gear bracket shall be of suitable length to accommodate the bonnet nut, bracket, and bracket securing nut. The performance of the valve shall be satisfactory when the hand lifting gear is used. The hand lifting gear is used within 100 lb/in^2 of operating pressure.

3.3.4 The nominal size of a safety valve shall be the initial inside diameter of the inlet opening to the valve. The sizes and approximate capacities of the valve shall be as specified (see 6.2.1). Ordinarily, valves will be ordered on the basis shown in table I.

TABLE I. Nominal sizes.

Type	Required capacity	Capacity tolerances	Nominal size
	(Lb/h)	(Lb/h)	(Inches)
I	60,000 maximum[1/]	-0, +5,000	2-1/2
III	55,000 maximum[1/]	-0, +5,000	2-1/2
I and III	30,000 maximum	-0, +5,000	1-1/2
II	20,000 maximum[1/]	-0, +2,500	1-1/2

[1/] 68,500 pounds per hour (lb/h) for type I and type III valves, and 24,000 lb/h for type II valves permitted on boilers having a total steam capacity in excess of 250,000 lb/h at full power.

3.3.5 Design pressure. Unless otherwise specified (see 6.2.1), safety valves shall be furnished in one of the standard design pressures specified in 3.3.5.1 or 3.3.5.2.

3.3.5.1 600 lb/in^2 standard. Valves for use on boilers having drum operating pressures up to 675 lb/in^2 (for which the safety valve popping pressure may exceed 675 lb/in^2, but will not exceed 750 lb/in^2) shall be designed for a working pressure of at least 750 lb/in^2 and in the case of superheater valves, for an operating temperature of at least 875°F.

3.3.5.2 1500 lb/in^2 standard. Valves for use on boilers having drum operating pressures of 676 lb/in^2 to 1600 lb/in^2 (for which the safety valve popping pressure may exceed 1600 lb/in^2, but will not exceed 1700 lb/in^2) shall be designed for a working pressure of at least 1700 lb/in^2.

3.3.6 Threaded parts. Threaded parts used in the construction of safety valves shall comply with FED-STD-H28 and as specified in 3.3.6.1 through 3.3.6.4.

3.3.6.1 Unified national thread standards. Unless otherwise specified (see 6.2.1), screw threads shall be in accordance with the unified national thread series.

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3.3.6.2 Unified national coarse-thread series. Unless otherwise specified (see 6.2.1), threads for studs and nuts shall be in accordance with the national coarse-thread series, class 2A or class 2B, as applicable. Set studs shall be chosen so as to best assure permanency of installation.

3.3.6.3 Pipe threads. Pipe threads shall be in accordance with FED-STD-H28.

3.3.6.4 Thread projection. Bolt-studs and studs shall be sufficiently long to insure that, when nuts are screwed home, there shall be at least one-half thread projection clear from the outer face of the nut. The threaded portion on such bolts and ends of studs shall be not less than one and one-half times the bolt or stud diameter.

3.3.7 Locking device. The valve cap shall be locked with a padlock.

3.3.8 Welding and fabrication. Welding shall be in accordance with MIL-STD-278. Fabricated assemblies shall be stress relieved as units prior to finish machining. The stress relieving shall be done in accordance with MIL-STD-278.

3.3.9 Shock. When shock testing is required (see 6.2.1), the valve shall not be damaged or malfunction when tested as specified in 4.6.8.

3.4 Detail design.

3.4.1 Relieving capacities. The relieving capacity of safety valves shall be as specified (see 6.2.1) and shall be based upon 103 percent of popping pressure.

3.4.1.1 The aggregate relieving capacity of the safety valves of each boiler shall be such that all of the steam generated at the overload condition will be discharged without allowing the drum pressure to increase more than 3 percent above the highest pressure at which any valve is set, and in no case to increase more than 6 percent above the boiler design pressure. The relieving capacity of the superheater safety valve shall be such that the superheater will not be damaged by overheating or bursting should the firing rate be maintained for 30 seconds at any firing rate from the specified minimum to maximum overload rate. The capacity of the steam drum safety valves shall be not less than 75 percent of the steam generated at the overload condition. The relieving capacity of the drum pilot actuator valve shall be as small as practicable consistent with proper design and operation.

3.4.2 Safety valve settings. The popping pressures of safety valves shall be determined as follows: The steam drum pilot actuator valve shall be set to pop and open the superheater safety valve at a pressure of 8 percent greater than the highest working steam drum pressure (120 percent overload). The first, second, third, and fourth steam drum safety valves shall be set to pop at pressures of 2, 3.5, 5, and 5 percent, respectively, of the highest working steam drum pressure higher than the pilot actuator valve setting. If the superheater safety valve is of the spring loaded type, it shall be set at a pressure of 1.5 percent of the highest working

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steam drum pressure higher than the pilot actuator valve setting. After calculating settings, round off to the nearest 5 lb/in^2 gage pressure to specify set pressures. If the superheater safety valve and the pilot actuator have the same setting after rounding off, then specify the superheater safety valve setting to be 5 lb/in^2 less than the pilot actuator setting. Prior to delivery, the popping and reseating pressures shall be set on all safety valves with steam pressure, within the maximum tolerances listed:

(a) Up to 750 lb/in^2 boiler drum pressure (600 lb/in^2 class valves) plus or minus 5 lb/in^2.
(b) 750 to 1500 lb/in^2 boiler drum pressure (1500 lb/in^2 class valves) plus or minus 10 lb/in^2.

Valves shall be tested (see 4.6.7) to determine acceptability of popping and reseating variation.

3.4.3 Springs. Springs shall be completely exposed to view. The permanent set of the springs, 10 minutes after being released from a cold compression test which closes the spring solid, shall not exceed 0.030 inch. The springs shall not be compressed solid when the valve is at its maximum lift (see 4.6.1).

3.4.3.1 The springs shall be such that the popping pressure can be varied over a range of approximately plus or minus 5 percent of the specified setting (see 4.6.4). The range of popping pressures for which the springs may be used satisfactorily shall be specified on the drawings (see 3.8.1).

3.4.3.2 Springs shall be protected from deterioration due to corrosion by a protective coating acceptable to the command or agency concerned. The type of coating shall be specified on the working drawings (see 3.8.1).

3.4.4 Blowdown adjusting device. The blowdown adjusting device shall remain symmetrical with the axis of the valve while the adjustment is being changed. A stop shall be provided for fixing the position of the device after adjustment is made.

3.4.4.1 The blowdown setting shall be adjustable without dismantling the valve.

3.4.4.2 Safety valves shall operate satisfactorily at any blowdown setting within the specified range (2-1/2 to 6 percent of popping pressure) without chattering. Valve closure shall be clean and sharp.

3.4.5 Valve nozzles (seat bushings) and discs. Valve nozzles shall be welded into the valve body or be of the "full nozzle" or "through bushing" type.

3.4.5.1 If the valve disc is guided from beneath, the guide rings shall extend below the seat to a distance not less than 1/2 of the valve diameter. In any case, guides shall be supported in such a way that lateral steam thrust while the valve is blowing cannot cause binding or seizing of the valve.

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3.4.6 Fittings. Fittings shall be provided as specified in 3.4.6.1 through 3.4.6.4.

3.4.6.1 A lock mechanism shall be provided which will prevent changing the spring adjustment without opening the lock.

3.4.6.2 Spring loaded safety valves shall be provided with a lever for the attachment of a hand lifting gear. Each valve shall have incorporated in this lifting mechanism, a device to prevent lifting the valve by hand beyond the maximum designed lift of the valve in order to prevent bending of the valve stem. The lifting gear shall be capable of 360 degree rotation.

3.4.6.3 Provisions shall be made in all valves for clamping the valve to its seat by means of a yoke or gag when testing the valve (or setting a valve of greater pressure). A tightening screw shall be provided with a knurled head or "wing nut" type of head to discourage the use of wrenches thereon. Casting and machining of yoke or gag and valve fitting which receives it shall be of sufficient accuracy to prevent lateral deflection or bending of valve spindle when the gag screw is tightened.

3.4.6.4 A lift indicator shall be furnished with each valve which shall indicate, by means of a pointer and a graduated scale, the amount of lift when the valve is popping. The scale shall be graduated to show zero lift, full lift, and intermediate lifts.

3.4.7 Universality. Valve bodies shall be such that the valve adjusting ports, drain connections, and actuator connection (type II) may be located or interchanged at any of these locations in the horizontal plane: 90 degrees, 180 degrees, and 270 degrees from the exhaust flange. These three locations shall have raised bosses.

3.4.7.1 Drain connection. A flanged 1/2-inch iron pipe size (i.p.s.) drain connection shall be added to the drain boss as shown on Drawing 5000-S5103-841622.

3.4.7.2 Adjusting ports. Adjusting ports shall be located so as to facilitate blowdown adjustment when the valves are installed on the boiler.

3.4.7.3 Actuator connection. For type II valves, a flanged 1/2-inch i.p.s connection shall be added to one of the three bosses.

3.4.8 Flanges. Unless otherwise specified (see 6.2.1), flanges (inlet, outlet, drain, and actuator line) shall conform to ANSI B16.5 for the applicable pressure at the flange. Flange finish shall be described on the applicable drawings (see 3.8.1). Flanges other than ANSI standard shall be specified only for replacements for safety valves originally not having ANSI standard flanges.

3.5 Interchangeability. Identical parts, including repair parts, of corresponding valves furnished on the contract or order or built to the same drawings, shall be physically and functionally interchangeable without machining or handfitting of any kind.

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3.6 Markings. Safety valve shall be plainly marked by the manufacturer. The markings may be stamped or cast on the body, or stamped on an identification plate permanently secured thereto and which shall be in accordance with MIL-P-15024 and MIL-P-15024/5. Identification plates shall contain the following information:

(a) Name or identifying trademark of manufacturer and designation number or symbol.
(b) The nominal size and nominal operating pressure for which the valve body is suitable (preferably cast on the valve body, as for example, 2-1/2 inches, -600).
(c) The authorized popping pressures.
(d) The authorized reseating pressure.
(e) The rated relieving capacity (as specified in 3.4.1.1).
(f) Lift of valve necessary to give rated relieving capacity.

3.7 Tools. The number and type of tools required to be furnished with each contract or order shall be as specified (see 6.2.1). These tools will normally consist of the following:

(a) One complete set of seat and disc gages or templates, as applicable for each size and type safety valve installed.
(b) Safety valve gags or testing yokes in the number specifically called for in the contract or order.
(c) Special tools and wrenches required for the proper maintenance of the type of valve. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the offices of the Defense Contract Administration Service Management Area (DCASMA)).

3.8 Technical data. The contractor shall prepare the following technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2), and as specified in 3.8.1 through 3.8.3.

3.8.1 Drawings. The contractor shall prepare a drawing which includes a list of material of valves in accordance with the data ordering document (see 6.2.2). Unless otherwise specified (see 6.2.1), all drum safety valves (type I, class 1) shall conform to the dimensions shown on Drawing 5000-S5103-841622, except as modified by 3.4.8. Other types of valves shall also conform to these dimensions insofar as practicable. When the inherent design of a valve is such that it cannot conform to these dimensions, and following conclusion of the tests specified in 4.2, the manufacturer shall furnish the command or agency concerned an outline drawing (tracing) of the valve, showing all limiting external dimensions and connections. Valves of this general type shall conform to this dimensioned master drawing.

3.8.1.1 The following statement shall appear on all applicable drawings: "This drawing shall not be released outside the Government or be used or disclosed in whole or in part for manufacture or acquisition without the written permission of the manufacturer, except for emergency repair or overhaul work by or for the Government."

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3.8.2 Information on drawings shall include the necessary runouts, parallelism of surfaces and concentricities. In addition, checks to be made during assembly of parts shall be listed.

3.8.3 Certification data sheets. Certification data sheets shall include lift data (actual, minimum, and maximum) table, spring data table, table of travel of lever and force required to raise disc 3/32 inch above seat at 50 percent boiler pressure, and schematic showing outlet, drain, and actuating flanges and set screw locations for all boilers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA). Qualification shall consist of the examination of 4.5 and tests specified in 4.6.1 through 4.6.3.1 and 4.6.4 through 4.6.8.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination in 4.5 and tests specified in 4.6.1, 4.6.2, 4.6.3, and 4.6.3.2.

4.4.1 Lot. Valves of the same type and class presented for delivery at one time shall be considered a lot.

4.4.2 Sampling for examination. A random sample of assembled valves shall be selected from each lot in accordance with table II for the examination specified in 4.5 through 4.5.1.2.

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TABLE II. Sampling for examination.

Number of valves in lot	Number of valves in sample	Acceptance number (defectives)
15 and under	7	0
16 to 25	10	0
26 to 40	13	0
41 to 65	16	0
66 to 110	21	0

4.4.3 Sampling for tests. Each valve in the lot shall be tested as specified in 4.4.

4.5 Examination. Each of the sample valves selected in accordance with 4.4.2 shall be visually and dimensionally examined to determine conformance with the requirements of this specification as to material, workmanship, and dimensions specified on the drawings. Material examination shall be as specified in 4.5.1 through 4.5.1.2.

4.5.1 Material certification reports (see 6.2.2) for pressure containing parts shall be examined to determine conformance with the applicable material specification.

4.5.1.1 The DCASMA will normally accept certification that non-pressure containing parts conform to their applicable material specification providing the certification contains actual test results. Certification representing stocks more than one year old are not acceptable.

4.5.1.2 When there is a question relative to material identification or quality of a lot, the DCASMA may request chemical analysis and mechanical tests as he deems necessary to establish complete identity and overall quality, using MIL-STD-271, where applicable.

4.6 Tests.

4.6.1 Permanent set of valve springs. The permanent set of the spring shall not exceed 0.030 inch and shall be measured 10 minutes after being released from a cold compression which has closed the spring solid (see 3.4.3).

4.6.2 Hydrostatic test. Valves shall be held closed by placing a gag on the valves "hand tight" (see 3.4.6.3) and then subjected to a water pressure 1-1/2 times the popping pressure of the valves. This test shall be used specifically for testing the integrity of the nozzle and disc insert and shall not be used for testing seat tightness. There shall be no leakage between the nozzle (seat bushing) and disc or through the body of the valve due to faulty castings. Water temperature shall not exceed 100°F. After the hydrostatic test, the valve will be opened and the spindle checked to ensure it is within manufacturer's tolerances.

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4.6.3 Steam tightness. Valves tested shall be free from preservative prior to testing. The contractor may add any tests or tightness requirements to ensure the maximum steam tightness. The valve shall be placed on a test stand or accumulator, and after a warm-up period, will be set to pop at the required pressure within the tolerances specified in 3.4.2 for the valve concerned. After setting, gag hand tight, then back off.

4.6.3.1 Steam tightness (qualification inspection). A steam tight cover shall be affixed to the exhaust flange and actuating flange, as applicable, and line connected to the drain flange. The drain line shall be connected to a cooling condenser. The condensate from the condenser shall be collected for measurement. Measurement shall be conducted as follows:

(a) Adjust steam pressure to 92 percent of set pressure.
(b) Measure the condensate from the above system for a period of 30 minutes.

Steam tightness acceptance shall be based upon the following:

(c) No visible steam leakage between the valve spindle and its guide.
(d) The leakage rate shall be not greater than the following:
    (1) 300 cubic centimeters per hour (cm^3/h) of condensate for 1-1/2 inch valve.
    (2) 500 cm^3/h of condensate for 2-1/2 inch valve.

4.6.3.2 Steam tightness (quality conformance inspection). The pressure shall be adjusted to 92 percent of set pressure. After removing piping from valve outlet flange, the valve shall be allowed to dry out, minimum drying out period 5 minutes, maximum 10 minutes. Visual examination shall be the standard for determining steam tightness. The open outlet of the valve is to be viewed against a dark background. The examining agent shall be stationed from 4 to 5 feet away from the outlet and at an angle of 45 degrees from the center line of the valve outlet. The valve shall be considered tight if there is no visible sign of escaping steam at the time of examination.

4.6.4 Range of popping pressure. Each valve shall be popped at three popping pressures, to cover the designed setting ranged of the spring (see 3.4.3.1). The setting shall be chosen such as to assure that the designed range of setting can be obtained. Simmering (slight constant escape of steam between the seat and disc) may be permitted prior to lifting. At all times, valve closure shall be clean and sharp.

4.6.5 Inspection of castings and welding. Any portion of the valve which is subject to full boiler pressure while in service shall be inspected in accordance with MIL-STD-278. Welding shall meet the requirements of MIL-STD-278.

4.6.6 Relieving capacity. The actual relieving capacity of the valves at the designed popping pressure plus 3 percent overpressure shall be measured to determine compliance with 3.4.1, and all capacities shall be determined on saturated steam with the reduction in capacity at normal service temperature being calculated for the superheater valves.

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4.6.7 Popping and reseating variation. Valves shall be tested as specified in 4.6.7.1 through 4.6.7.3 for compliance with the allowable variations in popping and reseating pressures.

4.6.7.1 Type I valves. Valves shall be set to pop and reseat at the designed popping and reseating pressures when the boiler is steaming at minimum steaming rate. After such setting, the boiler shall be brought up to the rating necessary to hold the valve open, and held for not more than one hour before popping each valve. Each valve shall then be popped with the boiler under running load (steam flowing through the main steam outlet) held open for 2 minutes and then reseated. The popping and blowdown pressure under this condition shall not vary from the designed settings by more than plus or minus 1 percent of the designed reseating. The valve shall be popped again and allowed to reseat with the time interval before popping and reseating being only that required to control the boiler pressure. The variations in popping pressure under this condition from the designed popping pressure shall not exceed 1-1/2 percent of the designed reseating pressure, and the variation of the reseating pressure from the designed reseating pressure shall not exceed 1 percent of that pressure. Any reduction in blowdown within the above tolerances shall not affect the proper functioning of the valve.

4.6.7.2 Type II and type III valves. In the case of pilot operated valve systems, the popping and reseating pressures of type II and type III valves shall be set initially under static load in the same general manner as specified for type I valves in 4.6.7.1. Having set the type II and type III valves in this manner, the steaming rate of the boiler shall be increased above the combined capacity of the drum pilot valve and the superheater outlet valves, and the steam temperature increased to designed operating temperature. This condition shall be maintained (as closely as possible) for at least 15 minutes before popping the valve. The pilot valve shall then be popped with the boiler under running load (steam flowing through the superheater outlet), held open for 2 minutes, and then reseated. Immediately after reseating, the valve shall be popped and allowed to reseat again, the time interval before popping and reseating being only that required to control the boiler pressure. The variation in popping and reseating pressures under either of the above conditions shall not exceed plus or minus 1 percent of the designed reseating pressure. Blowdown may be compensated an additional 1 percent for every 100°F of superheater differential temperature.

4.6.7.3 Time lag between pilot valve and superheater unloading valve. In addition to the tests specified in 4.6.7.1 and 4.6.7.2 (or in conjunction therewith), runs shall be made to ascertain the time lag between popping and reseating of the drum pilot valve and the lifting and reseating of the superheater valve. Under static load, with the main stop secured, the interval between popping of the pilot valve and lifting of the superheater valve shall not exceed 3 seconds. Similarly, the interval between reseating of the pilot valve and reseating of the superheater valve shall not exceed 3 seconds. Under running load with designed steam temperature at the superheater outlet, the time lag shall not exceed 2 seconds in either case.

4.6.8 Shock. When specified (see 6.2.1), one valve of each new design shall be tested in accordance with grade A, class I, type A requirements of MIL-S-901.

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4.7 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation-packaging, packing, and marking. Valves shall be individually preserved-packaged level A or C, packed level A, B, or C, and marked in accordance with MIL-V-3, as specified (see 6.2.1).

5.2 Tools. Tools shall be preserved-packaged, packed, and marked in accordance with MIL-R-196, level A, B, or C, as specified (see 6.2.1).

5.3 Cushioning and cushioning materials. Cushioning and cushioning materials, including the requirements for cushioning and dunnage (loose-fill) shall be in accordance with the requirements specified in MIL-R-196.

5.4 Data (see 6.2.2). Drawings, manuals, reports, etc. shall be packaged in a greaseproof, waterproof, heat-sealed bag.

6. NOTES

6.1 Intended use. The boiler safety valves covered by this specification are intended to vent excess steam, protecting the boiler from over-pressurization.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, and size required (see 1.2 and 3.3.4).
(c) If screw threads or threads for studs and nuts are other than specified (see 3.3.6.1 and 3.3.6.2).
(d) Whether shock test is required (see 3.3.9 and 4.6.8).
(e) Relieving capacity and popping and reseating pressure (see 3.4.1).
(f) If replacement valve, advise flange standard required (see 3.4.8).
(g) Number and type of tools required (see 3.7).
(h) Level of preservation-packaging, packing, and marking (see 5.1 and 5.2).
(i) Boiler operating pressure and whether boiler is constant drum pressure or constant superheater outlet pressure. If the latter, indicate highest working steam drum pressure (120 percent overload). Indicate superheater outlet temperature (120 percent overload).

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6.2.2 Data requirements. When this specification is used in a contract which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID	Option
3.3.9.1	Calculations	UDI-E-23213	------
3.8.1 through 3.8.3	Drawings, engineering and associated lists	DI-E-7031	Level 2 Design activity designation - Contractor Drawing number - Contractor Certification data sheets - Required
4.5.1	Certification data/report	UDI-A-23264	------

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List QPL 17462 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

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6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Cross-reference of classifications. Valves covered by this specification are equivalent to the valves of MIL-V-17462A(SHIPS) as follows:

This specification	MIL-V-17462A(SHIPS)
Type I	Type I, class 1
Type II	Type II, class 1
Type III	Type III, class 4
--------	Class 2
--------	Class 3
Class 1	----------
Class 2	----------
Size 1	Size 1
Size 2	Size 2

6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 Changes from previous issue. Asterisks (*) are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - SH
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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-V-17462B(SH)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

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